

Brac University
Department of Electrical & Electronic Engineering
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Course Number : EEE308L

Section :06

Group No :04



Lab Report

Experiment no: 05

Name of the experiment:

Study of Operational Amplifier as Active Filter.

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Objective:

The main objective of this experiment is to study op-amp as the lowpass and the highpass filters.

Theoretical Background:

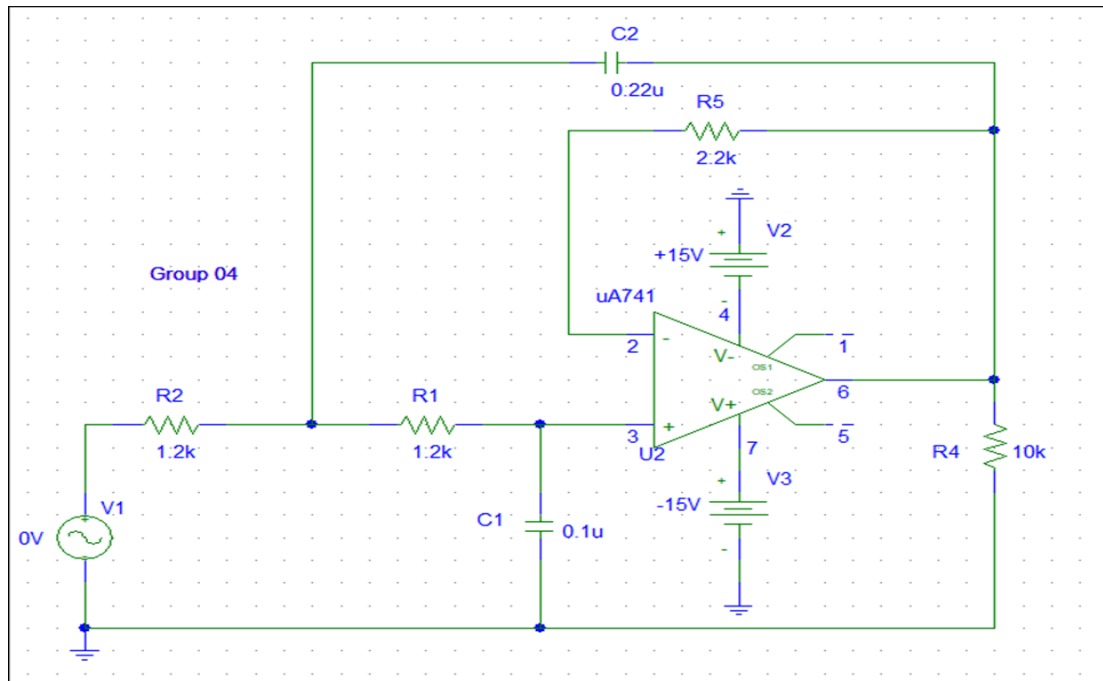
In this lab, we explored how active filters work, specifically focusing on the low-pass variety. This type of circuit is designed to let low-frequency signals through while filtering out higher frequencies. By using an op-amp with resistors and capacitors, we were able to build an effective filter without needing bulky inductors. We chose a Butterworth design to ensure the output remained smooth and stable within the passband. Since the circuit was set up for unity gain, the output stayed very close to the input level until reaching the cutoff frequency. By measuring the output across a range of frequencies, we identified the -3 dB cutoff point, after which the signal strength drops rapidly at a rate of 40 dB per decade.

Equipment:

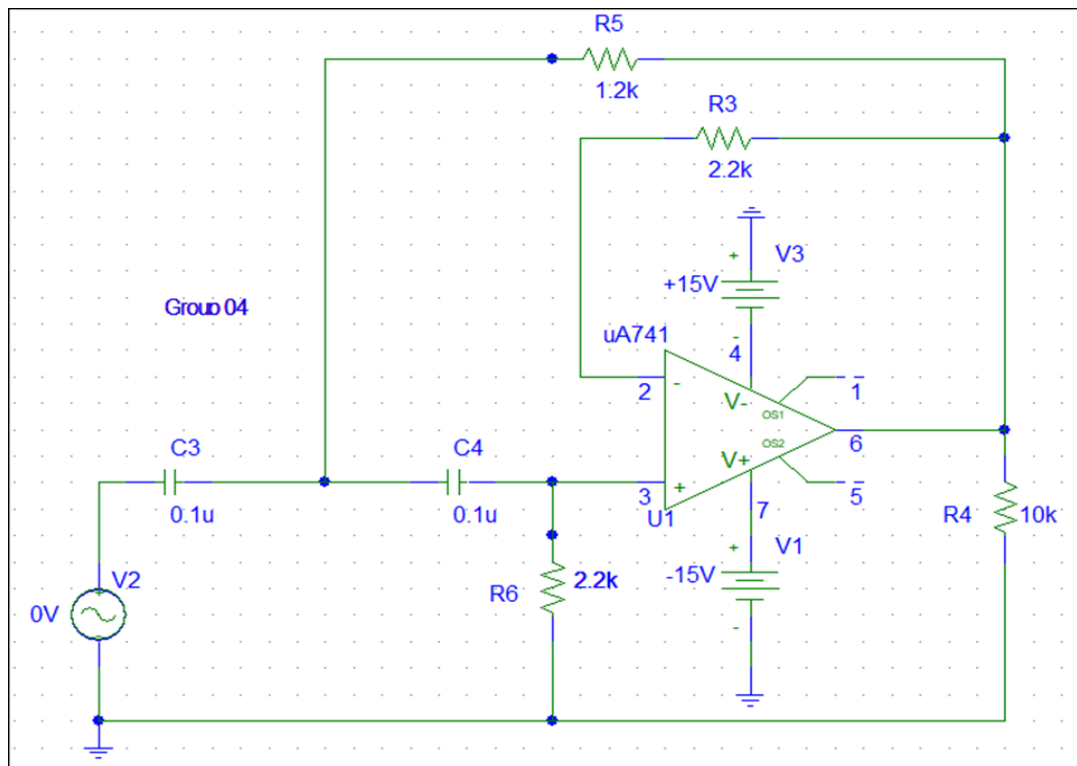
- Op-amp(741)
- Resistor ($9.9\text{ K}\Omega$)(2 pieces)
- Resistor ($2.2\text{ K}\Omega$)(2 pieces)
- Resistor ($2.19\text{ K}\Omega$) (1 pieces)
- Resistor ($1.2\text{ K}\Omega$)(2 pieces)
- Capacitor ($0.215\text{ }\mu\text{F}$)(2 pieces)
- Capacitor($0.1\text{ }\mu\text{F}$)(2 Pieces)
- DC power supply
- Oscilloscope
- Signal Generator

Circuit Diagram:

-40 dB / Decade Low Pass Butterworth Filter:



+40 dB / Decade High Pass Butterworth Filter:



Data Table:**-40 dB / Decade Low Pass Butterworth Filter:**

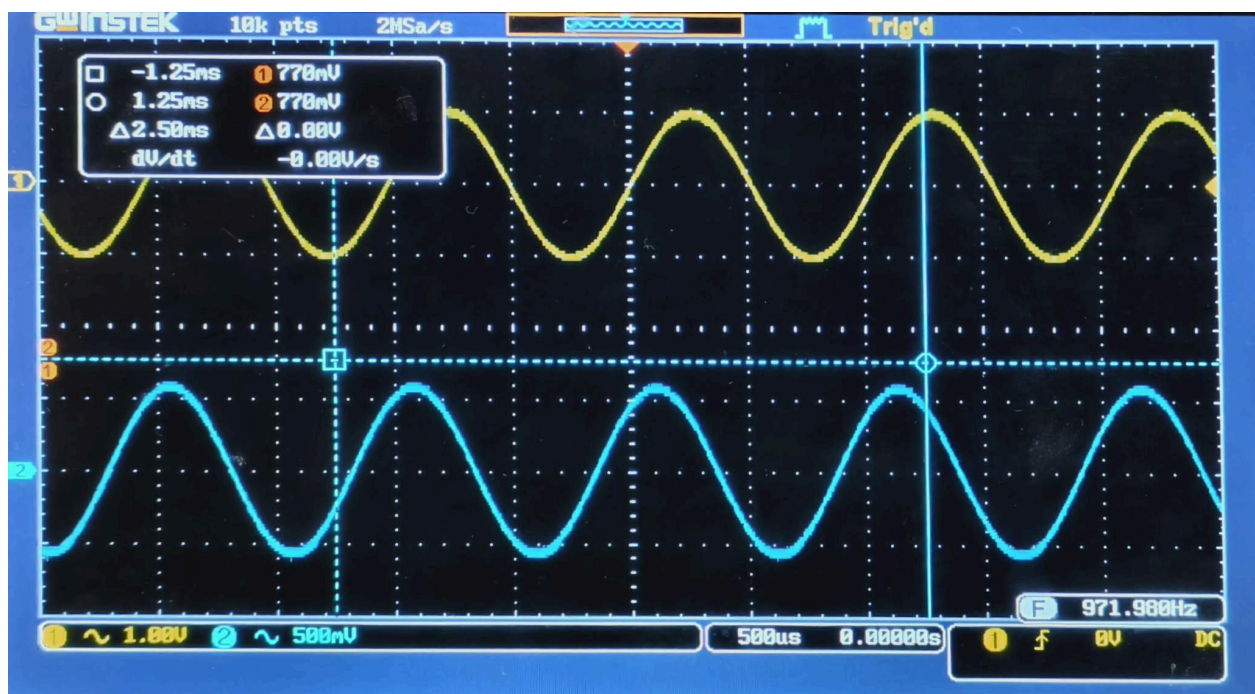
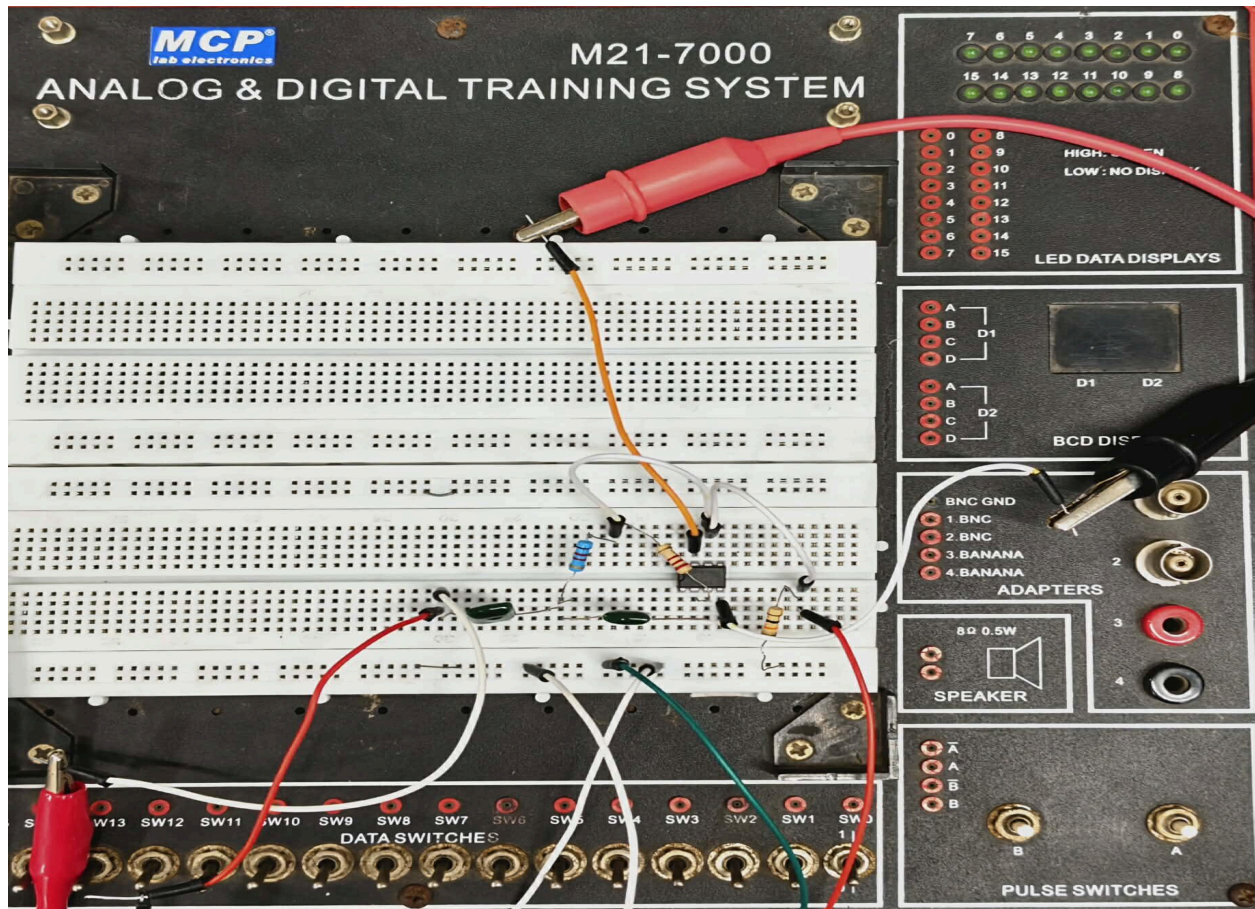
Serial No.	f(Hz)	V _{o,a} (V)	A _v (dB)= $20\log(\frac{V_{o,a}}{V_{i,a}})$
1	10	1.02	0.172
2	50	1.04	0.340
3	100	1.04	0.340
4	500	0.86	-1.31
5	845.92	0.70	-3.098
6	5k	0.16	-15.917
7	10k	0.11	-19.172

+40 dB / Decade High Pass Butterworth Filter:

Serial No.	f(Hz)	V _{o,a} (V)	A _v (dB)= $20\log(\frac{V_{o,a}}{V_{i,a}})$
1	100	0.08	-21.94
2	500	0.392	-8.134
3	972.09	0.66	-3.61
4	5k	1	0
5	10k	1.02	0.17
6	50k	1.02	0.17
7	100k	1.02	0.17

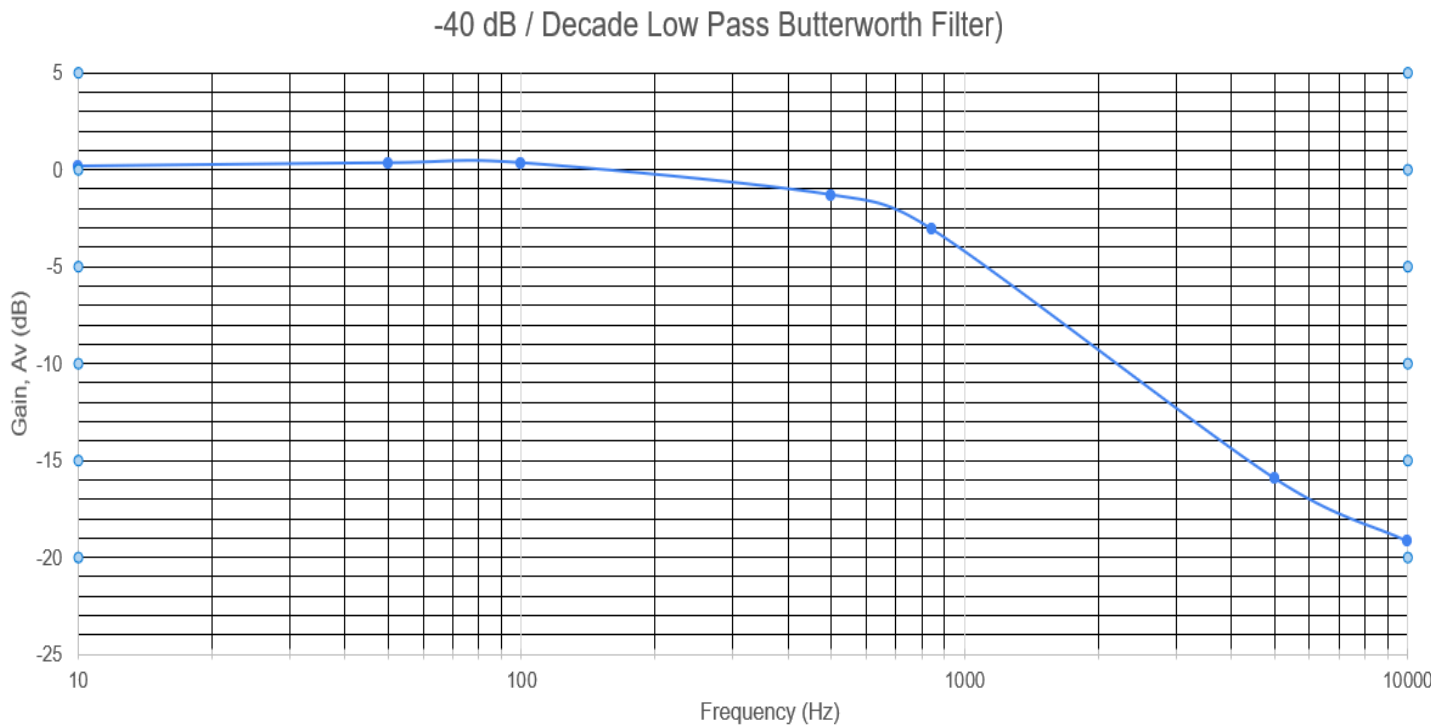
Hardware Setup:

+40 dB / Decade High Pass Butterworth Filter:

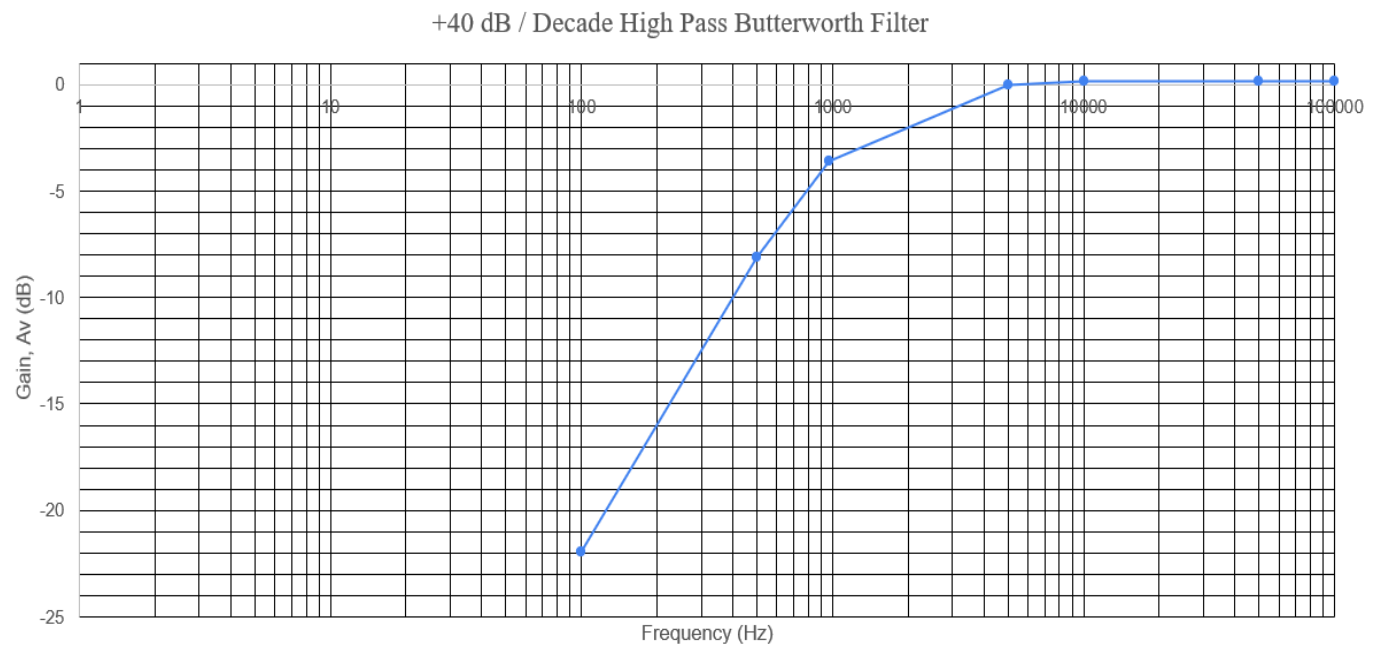


Characteristics Curve :

-40 dB / Decade Low Pass Butterworth Filter:



40 dB / Decade High Pass Butterworth Filter:



Discussion:

In this experiment, we successfully built and tested two types of active filters: a low pass and a high pass. For the low pass filter, we calculated a theoretical cutoff frequency of 937.6 Hz , while our actual measured results showed a cutoff at 845.92 Hz. The high pass filter performed similarly, with a measured cutoff frequency of 972.09 Hz. Both circuits followed the expected Butterworth design, providing a smooth output in the passband and a sharp drop-off of 40 dB per decade once the signal passed the cutoff point. The small differences between our calculated and measured values were likely due to the slight variations in the actual resistor and capacitor values used on the breadboard.

Appendix:-

□ -40dB/Decade Low Pass Butterworth Filter:

S/N	f(Hz)	$V_{o,a}(V)$	$A_{v,dB} = 20 \log \left(\frac{V_{o,a}}{V_{i,a}} \right)$
1	10	1.02	0.172
2	50	1.04	0.340
3	100	1.04	0.340
4	500	0.86	-1.31
5	1K 845.92 Hz	0.62 0.70	-4.152 -3.098
6	5K	0.16	-15.917
7	10K	0.11	-19.172

□ 40dB/Decade High Pass Butterworth Filter:

S/N	f(Hz)	$V_{o,a}(V)$	$A_{v,dB} = 20 \log \left(\frac{V_{o,a}}{V_{i,a}} \right)$
1	100	0.08	-21.94
2	500	0.392	-8.134
3	1K 972.09	0.66	-3.61
4	5K	1	0
5	10K	1.02	0.17
6	50K	1.02	0.17
7	100K	1.02	0.17

For Low Pass

Resistor values, $R_1 = 1.20 \text{ k}\Omega$

$$R_2 = 1.21 \text{ k}\Omega$$

$$R_f = 2.35 \text{ k}\Omega$$

$$R_L = 9.98 \text{ k}\Omega$$

Capacitor values, $C_1 = 0.241 \text{ }\mu\text{F}$

$$C_2 = 0.102 \text{ }\mu\text{F}$$

Theoretical cutoff; $f_c = 937.6 \text{ Hz}$

$$\begin{aligned}\text{Found cutoff; } f_{cp} &= \frac{1}{2\pi \sqrt{(1200)(1200)(0.241)(0.102) \times 10^{-6}}} \\ &= 845.92 \text{ Hz}\end{aligned}$$

For High Pass:

$$R_1 = 1.20 \text{ k}\Omega$$

$$C_1 = 0.102 \text{ }\mu\text{F}$$

$$C_2 = 0.100 \text{ }\mu\text{F}$$

$$R_2 = 2.19 \text{ k}\Omega$$

$$\begin{aligned}\text{Found cutoff; } f_{hp} &= \frac{1}{2\pi \sqrt{(1200 \times 2190) \times (0.102 \times 0.100) \times 10^{-6}}} \\ &= 972.09 \text{ Hz}\end{aligned}$$

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